HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained **predictive models** using **Python** and **scikit-learn** to support actuarial forecasting, achieving an average **RMSE of 0.78** on historical data.
- Engineered **data pipelines** with **Python** and **SQL** to automate the extraction, transformation, and loading of financial data for actuarial reporting, processing approximately **50 GB** of data daily.
- Designed and implemented **interactive dashboards** and **data visualisations** using **Power BI** and **DAX** to communicate complex analytical findings to actuarial and finance stakeholders, delivering **15 key reports** monthly.
- Collaborated with actuarial and risk stakeholders to gather requirements for analytical solutions, translating business needs into technical specifications for model development and data integration.
- Supported the enhancement of reporting processes through the development of **Python scripts** for data aggregation and anomaly detection, reducing manual reporting effort by an estimated **20%**.
- Performed statistical analysis and hypothesis testing using **R** and **pandas** to validate actuarial assumptions and assess the impact of market trends on financial risk.
- Containerized analytical applications using **Docker** and managed deployments on **AWS EC2** instances, ensuring scalable and reliable execution of data processing tasks.

SKILLS

Programming & Data Engineering – Python, SQL, R, pandas, NumPy, scikit-learn, PySpark

Machine Learning & Statistics – Predictive Modelling, Forecasting, Trend Analysis, Regression, Hypothesis Testing, Supervised Learning, Unsupervised Learning

Data Visualisation & BI – Power BI, Tableau, DAX, Power Query, Data Modelling

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, CI/CD, Jenkins, Git

Actuarial & Finance – Actuarial Reporting, Financial Risk Analysis, Assumption Validation, Surplus Analysis, Financial Decision-Making

PROJECTS

Actuarial Assumption Validation Framework

- Developed a Python framework to automate the validation of actuarial assumptions against historical data, incorporating statistical tests and trend analysis.
- Built a **Power BI dashboard** to visualise assumption deviations and historical performance, enabling actuaries to quickly identify areas requiring re-evaluation.
- Applied regression analysis using **scikit-learn** on a dataset of **~1.5 million** policy records to model claim frequencies, achieving an **R-squared of 0.82**.

Financial Risk Forecasting Model

- Engineered a **time-series forecasting model** using Python and pandas to predict key financial risk indicators, processing data from multiple enterprise systems.
- Trained a **Gradient Boosting Regressor** model on **~280,000** historical financial data points to forecast potential solvency shortfalls, achieving a **mean absolute error of 5.3%**.
- Integrated model outputs into a reporting suite, providing actionable insights to finance and actuarial teams for strategic planning.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)